# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

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| **Course Code:** | CSA10 / CSA1063 | **Course Title:** | Software Engineering |
| **CO Assessed:** | CO2 | **Assessment:** | Assessment Tool 2 – Scenario-Based Requirement Analysis |
| **Weightage:** | 20% | **Total Marks:** | 25 |
| **CO2 Statement:** | *Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills* | | |
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| **Date:** | 04-08-2026 | **Duration:** | 60 minutes |

### Code of Conduct

I A. Govardhan Reddy (192421359) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A. Govardhan Reddy

## Annexure A – Scenario-Based Requirement Analysis

### Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

# 0. Assigned Scenario / Problem Statement

A multi-specialty hospital currently manages patient registration, appointments, medical records, billing, pharmacy stock, and laboratory workflows using a combination of paper registers and disconnected spreadsheets. As patient volume has grown, this manual approach has led to long queues, duplicate or lost records, delayed lab-report delivery, billing errors, and poor visibility into bed and staff availability. The hospital management wants to commission a web-based Hospital Management System (HMS) that digitizes and integrates these workflows across the front desk, clinical staff, pharmacy, laboratory, and administration, while keeping patient data secure and the system easy to use for staff with limited computer literacy.

This document analyses the above scenario and presents the corresponding Software Requirement Specification.

# 1. Scenario Understanding and Problem Analysis

## 1.1 Business Domain

Healthcare services — specifically hospital administration and clinical operations for a multi-specialty hospital handling outpatient (OP) and inpatient (IP) care, diagnostics, pharmacy, and billing.

## 1.2 Stakeholders

- Patients — receive care, book appointments, and make payments.
- Doctors — diagnose, treat, and prescribe medication.
- Nurses — assist with ward care and bed management.
- Receptionists / Front-office staff — handle registration, appointment scheduling, and billing.
- Pharmacists — dispense medicines and manage stock.
- Lab Technicians — conduct diagnostic tests and publish reports.
- Hospital Administrators — oversee staff, resources, and reporting.
- Hospital Management (owners/board) — require operational and financial reports.
- External Payment Gateway / Insurance Providers — process payments and claims.

## 1.3 Existing Challenges

- Paper-based and spreadsheet-based records causing duplication, loss, and inconsistency.
- Long patient waiting times due to manual registration and appointment booking.
- Delayed communication of lab results between labs, doctors, and patients.
- Billing errors from manually compiled consultation, pharmacy, and lab charges.

- No real-time visibility into bed occupancy or pharmacy stock levels.
- Difficulty generating accurate administrative and statutory reports.

### 1.4 System Objectives

- Provide a single integrated platform for registration, scheduling, records, billing, pharmacy, and lab workflows.
- Reduce patient waiting time and manual paperwork.
- Ensure accurate, real-time, and auditable medical and financial records.
- Improve visibility of hospital resources (beds, staff, inventory) for better decision-making.
- Protect patient data confidentiality in line with healthcare data-privacy expectations.

## 2. Functional and Non-Functional Requirements

### 2.1 Functional Requirements (FR)

| ID | Functional Requirement | Description |
|---|---|---|
| FR-01 | Patient Registration | The system shall allow front-office staff or patients to register new patients with demographic and contact details. |
| FR-02 | Appointment Booking | The system shall allow patients/receptionists to book, reschedule, or cancel appointments with a chosen doctor and time slot. |
| FR-03 | Electronic Medical Records (EMR) | The system shall allow doctors and nurses to view and update patient medical history, diagnoses, and treatment notes. |
| FR-04 | e-Prescription | The system shall allow doctors to create digital prescriptions that pharmacists can view and dispense against. |
| FR-05 | Lab Test Ordering & Reporting | The system shall allow doctors to order lab tests and lab technicians to upload/publish results linked to the patient record. |
| FR-06 | Bed / Ward Management | The system shall allow nurses/admin staff to view and update real-time bed availability and patient admission/discharge status. |
| FR-07 | Billing & Invoicing | The system shall automatically consolidate consultation, pharmacy, and lab charges into a single itemized bill. |
| FR-08 | Payment Processing | The system shall support online and counter payments, integrating with an external payment gateway. |
| FR-09 | Pharmacy Inventory Management | The system shall track medicine stock levels, expiry dates, and trigger low-stock alerts. |
| FR-10 | Staff Scheduling | The system shall allow administrators to create and manage duty rosters for doctors, nurses, and support staff. |

| | | |
|---|---|---|
| FR-11 | Reporting & Analytics | The system shall generate administrative reports (occupancy, revenue, patient footfall) for management. |
| FR-12 | User Authentication & Role-Based Access | The system shall authenticate all users and restrict functionality based on role (patient, doctor, nurse, pharmacist, lab tech, admin). |
| FR-13 | Notifications | The system shall send appointment reminders and report-ready alerts via SMS/email. |

### 2.2 Non-Functional Requirements (NFR)

| Category | Requirement |
|---|---|
| Performance | The system shall respond to search/booking requests within 2 seconds under normal load (up to 500 concurrent users). |
| Security | Patient data shall be encrypted in transit (TLS) and at rest; access shall be role-based and all record access shall be logged for audit. |
| Reliability | The system shall have 99.5% uptime during hospital operating hours and support automated daily backups. |
| Usability | The interface shall be simple enough for staff with basic computer literacy to complete registration/billing within 3 clicks of navigation. |
| Scalability | The system shall support addition of new departments, doctors, and branches without major architectural change. |
| Maintainability | The system shall be modular so that individual modules (billing, pharmacy, lab) can be updated independently. |
| Availability | Core patient-care modules (registration, EMR, appointments) shall be available 24×7, including during scheduled maintenance of non-critical modules. |
| Interoperability | The system should support standard data-export formats (CSV/PDF/HL7 where applicable) for lab and insurance interfaces. |

## 3. Actors and Use Cases

### 3.1 Primary Actors

- Patient — initiates registration, appointment booking, and payment.
- Doctor — accesses records, prescribes medication, orders lab tests.
- Nurse — manages beds and updates patient care notes.
- Receptionist — registers patients, books appointments, generates bills.

- Pharmacist — dispenses medicine and manages inventory.
- Lab Technician — processes and publishes lab test results.
- Administrator — manages staff schedules and generates reports.

### 3.2 Secondary Actors

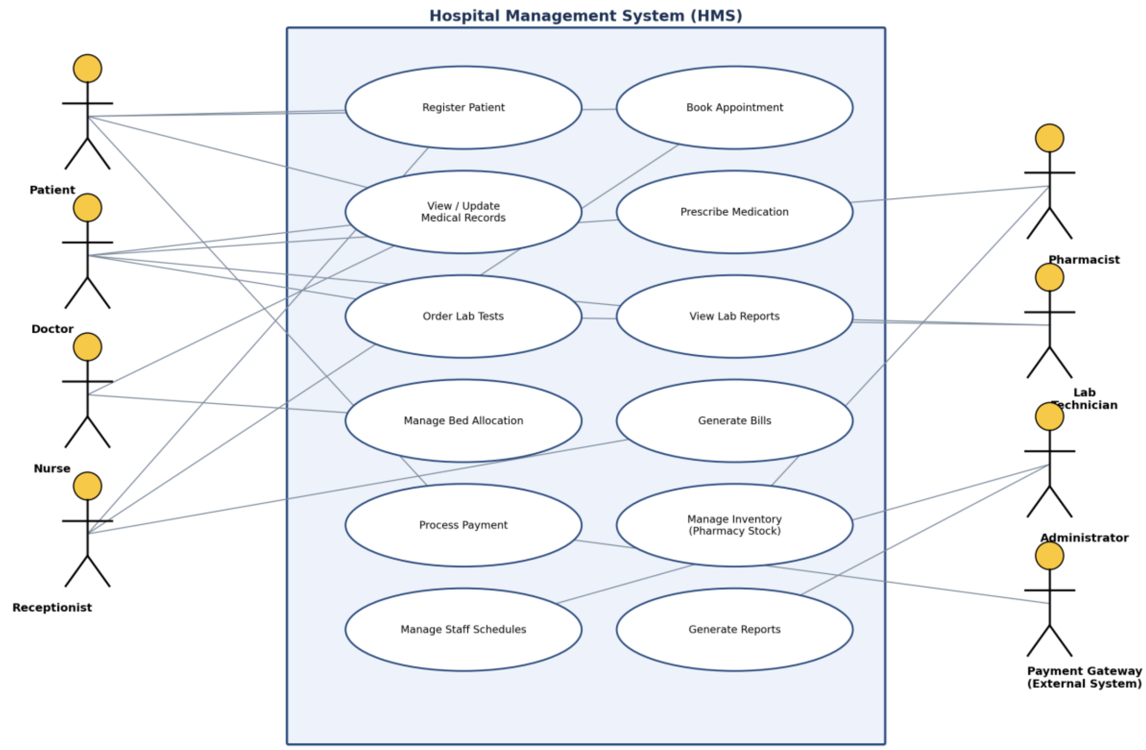
- Payment Gateway (external system) — processes online payments.
- SMS/Email Notification Service (external system) — delivers alerts and reminders.

### 3.3 Major Use Cases by Actor

| Actor | Associated Use Cases |
| --- | --- |
| Patient | Register Patient, Book Appointment, View Medical Records, Process Payment |
| Doctor | View/Update Medical Records, Prescribe Medication, Order Lab Tests, View Lab Reports |
| Nurse | Manage Bed Allocation, View/Update Medical Records |
| Receptionist | Register Patient, Book Appointment, Generate Bills |
| Pharmacist | Prescribe Medication (dispense), Manage Inventory |
| Lab Technician | Order Lab Tests (fulfil), View Lab Reports (publish) |
| Administrator | Manage Staff Schedules, Generate Reports |
| Payment Gateway | Process Payment |

### 3.4 Use Case Diagram

The diagram below represents the interactions between the identified actors and the Hospital Management System, following UML use-case notation (system boundary box, ellipses for use cases, stick figures for actors, and association lines).




# 4. Software Requirement Specification (SRS)

## 4.1 Introduction

This SRS defines the requirements for the Hospital Management System (HMS), a web-based platform intended to digitize and integrate patient registration, appointment scheduling, electronic medical records, pharmacy, laboratory, billing, and administrative reporting for a multi-specialty hospital.

## 4.2 Problem Description

The hospital currently relies on manual and disconnected record-keeping, causing delays, errors, and poor visibility into resources (see Section 1). The HMS is intended to resolve these issues through a single integrated software platform.

## 4.3 Scope of the System

- In scope: patient registration, appointment scheduling, EMR, e-prescription, lab order/result management, bed management, billing and payments, pharmacy inventory, staff scheduling, notifications, and reporting.
- Out of scope: telemedicine/video consultation, insurance claim adjudication logic, and ambulance/fleet management (may be considered in future phases).

## 4.4 Functional Requirements

Refer to Section 2.1 (FR-01 to FR-13) for the complete list of functional requirements.

## 4.5 Non-Functional Requirements

Refer to Section 2.2 for performance, security, reliability, usability, scalability, maintainability, and availability requirements.

## 4.6 Use Case Descriptions

Two representative use cases are elaborated below; the remaining use cases follow the same template.

| Field | Details |
|---|---|
| Use Case | UC-02: Book Appointment |
| Actor(s) | Patient (primary), Receptionist (alternate initiator) |
| Precondition | Patient is registered in the system. |
| Main Flow | 1. Patient/Receptionist selects department and doctor. 2. System displays available slots. 3. Patient/Receptionist selects a slot and confirms. 4. System reserves the slot and sends a confirmation notification. |
| Alternate Flow | If no slot is available, the system suggests the next available slot or an alternate doctor. |
| Postcondition | Appointment is created and visible in the doctor's schedule; confirmation notification is sent. |

| Field | Details |
|---|---|
| Use Case | UC-05: Order Lab Tests & View Reports |
| Actor(s) | Doctor (initiator), Lab Technician (fulfils) |
| Precondition | Patient has an active consultation/EMR entry. |
| Main Flow | 1. Doctor selects required tests from the EMR screen. 2. System routes the order to the laboratory queue. 3. Lab Technician records results and uploads the report. 4. System links the report to the patient's EMR and notifies the doctor. |
| Alternate Flow | If a sample is rejected, the Lab Technician flags it and the system notifies the ward/reception to re-collect the sample. |
| Postcondition | Lab report is stored in the EMR and accessible to the doctor and patient. |

## 4.7 Data Requirements

- Patient data: demographics, contact details, medical history, allergies, visit records.
- Clinical data: diagnoses, prescriptions, lab orders and results, admission/discharge records.

- Administrative data: staff schedules, bed occupancy, department/doctor master data.
- Financial data: billing line items, payment transactions, invoices.
- Inventory data: medicine stock, batch/expiry information, reorder thresholds.
- All personally identifiable and medical data must be stored with access controls and retained per applicable healthcare record-retention norms.

### 4.8 External Interface Requirements

- User Interfaces: responsive web portal for staff; simplified patient-facing portal/kiosk for registration and appointment booking.
- Hardware Interfaces: barcode/RFID scanners for pharmacy stock (optional), receipt/bill printers.
- Software Interfaces: integration with an online Payment Gateway API and an SMS/Email Notification Service API.
- Communication Interfaces: HTTPS for all client-server communication; secure REST APIs for third-party integrations.

### 4.9 System Features

- Role-based dashboards tailored to each actor (patient, doctor, nurse, pharmacist, lab technician, receptionist, administrator).
- Centralized searchable patient record accessible across departments.
- Automated consolidated billing across consultation, pharmacy, and lab charges.
- Real-time bed and inventory availability indicators.
- Configurable notification triggers (appointment reminders, low-stock alerts, report-ready alerts).
- Audit trail of record access and modifications for compliance.

## 5. Assumptions and Constraints

### 5.1 Assumptions

- Hospital staff will have basic digital literacy and access to desktop/tablet devices.
- A stable internet connection is available across hospital departments.
- Patients consent to electronic storage and processing of their medical data.
- Existing paper records will be migrated gradually and are not required to be bulk-imported on day one.

### 5.2 Constraints

| Type | Constraint |
|---|---|
| Technical | The system must integrate with the hospital's existing network infrastructure and any legacy diagnostic equipment interfaces. |

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| Operational | The system must support uninterrupted hospital operations during rollout; downtime must be scheduled outside peak OP hours. |
| Economic | Development and licensing costs must fit within the hospital's approved IT budget; open-source components are preferred where feasible. |
| Legal | The system must comply with applicable healthcare data-protection and medical-record-retention regulations. |
| Time-related | The system must be delivered and piloted in one department within the agreed project timeline before hospital-wide rollout. |

# 6. Validate Requirement Completeness and Consistency

## 6.1 Stakeholder Coverage Check

Each stakeholder identified in Section 1.2 maps to at least one functional requirement and use case: Patients (FR-01, FR-02, FR-08), Doctors (FR-03, FR-04, FR-05), Nurses (FR-03, FR-06), Receptionists (FR-01, FR-02, FR-07), Pharmacists (FR-04, FR-09), Lab Technicians (FR-05), Administrators (FR-10, FR-11), and the Payment Gateway (FR-08). This confirms all stakeholder needs identified during scenario analysis have been addressed.

## 6.2 Ambiguity, Redundancy, and Inconsistency Check

- Ambiguity: Terms such as "real-time" (FR-06) and "secure" (Security NFR) have been quantified where possible (e.g., 2-second response, TLS encryption) to avoid vague interpretation.
- Redundancy: Billing-related needs raised separately by receptionists and patients were consolidated into a single requirement (FR-07/FR-08) rather than duplicated.
- Inconsistency: FR-05 (lab ordering) and the Use Case Diagram were cross-checked to ensure the Doctor and Lab Technician roles are consistently represented in both artifacts.
- Missing requirements identified during review: an explicit user-authentication and role-based access requirement (FR-12) and a notification requirement (FR-13) were added after the initial pass, since they were implied but not stated by stakeholders.

## 6.3 Suggested Improvements

- Add measurable acceptance criteria to each NFR (e.g., exact concurrent-user targets validated through load testing) before development sign-off.
- Conduct a formal requirements walkthrough with a representative from each stakeholder group to validate priority and catch domain-specific gaps (e.g., insurance-claim nuances).

- Introduce a requirements traceability matrix linking each FR/NFR to its originating stakeholder need, use case, and test case.
- Prioritize requirements (e.g., MoSCoW method) to align the phased rollout with the time and budget constraints identified in Section 5.2.